

Unit 5 – Sampling Distributions
HW 4 – Sampling Distribution of Sample Means

Name: _____

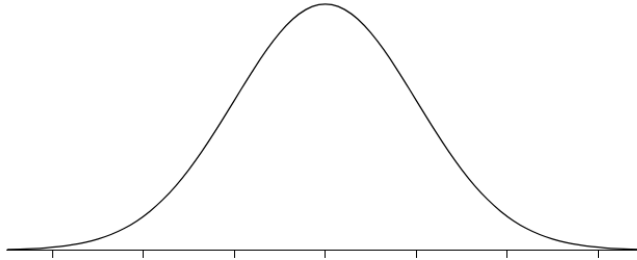
Directions: For the following parameters and statistics, find the sampling distribution, sketch the curve of the sampling distribution, and find the probability listed (show the work for your z-score!).

1) $\mu = 50$
 $\sigma = 18$
 $n = 36$

$\mu_{\bar{x}} = \underline{\hspace{2cm}}$

$\sigma_{\bar{x}} = \underline{\hspace{2cm}}$

$P(\bar{x} < 52) =$

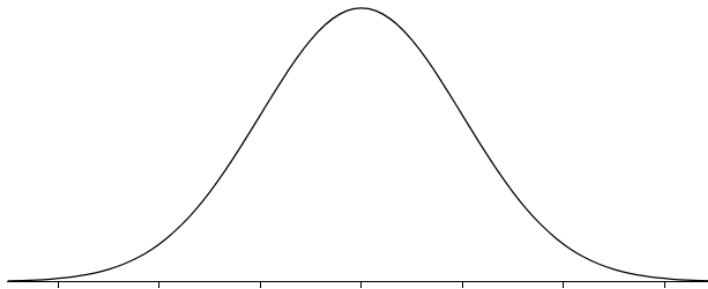


2) $\mu = 70$
 $\sigma = 14$
 $n = 16$

$\mu_{\bar{x}} = \underline{\hspace{2cm}}$

$\sigma_{\bar{x}} = \underline{\hspace{2cm}}$

$P(\bar{x} \geq 64) =$

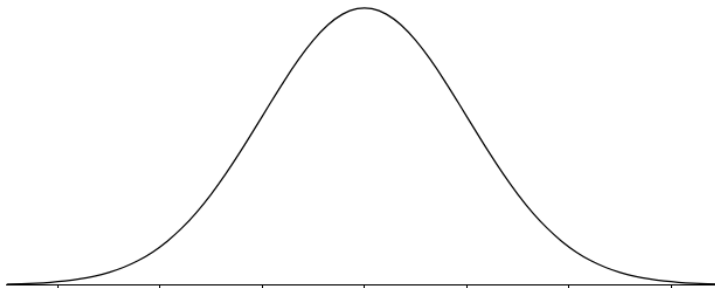


3) $\mu = 550$
 $\sigma = 29$
 $n = 121$

$\mu_{\bar{x}} = \underline{\hspace{2cm}}$

$\sigma_{\bar{x}} = \underline{\hspace{2cm}}$

$P(\bar{x} \leq ?) = 0.25$

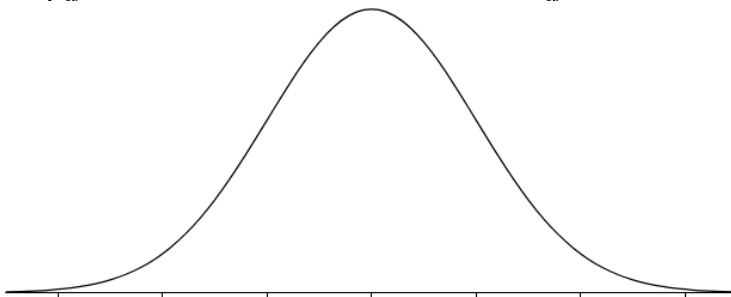


4) $\mu = -1$
 $\sigma = 4$
 $n = 64$

$\mu_{\bar{x}} = \underline{\hspace{2cm}}$

$\sigma_{\bar{x}} = \underline{\hspace{2cm}}$

$P(-1.75 < \bar{x} < -0.22)$



5) In Chicago, the average commute time is known to follow a normal distribution with a mean of 25 minutes and a standard deviation of 5 minutes. Suppose we take a random sample of 100 commuters from Chicago.

(a) Find and interpret the following: $P(X \leq 23)$

(b) Find and interpret the following: $P(\bar{x} \leq 23)$

(c) Explain why are the probabilities in part (a) and part (b) different.

6) A coffee company is conducting a taste test for its new blend of coffee beans. The caffeine content in their new blend follows a normal distribution with a mean of 90 milligrams and a standard deviation of 5 milligrams. To test the quality of the new blend, the company randomly selects a sample of 35 cups of coffee made from this blend. What is the probability that the average caffeine content in this sample of 35 cups is more than 92 milligrams? *Show ALL work: Communicate, Conditions, Calculate, and Conclude.*